Angular Manifold Routing: Plain■Language Overview

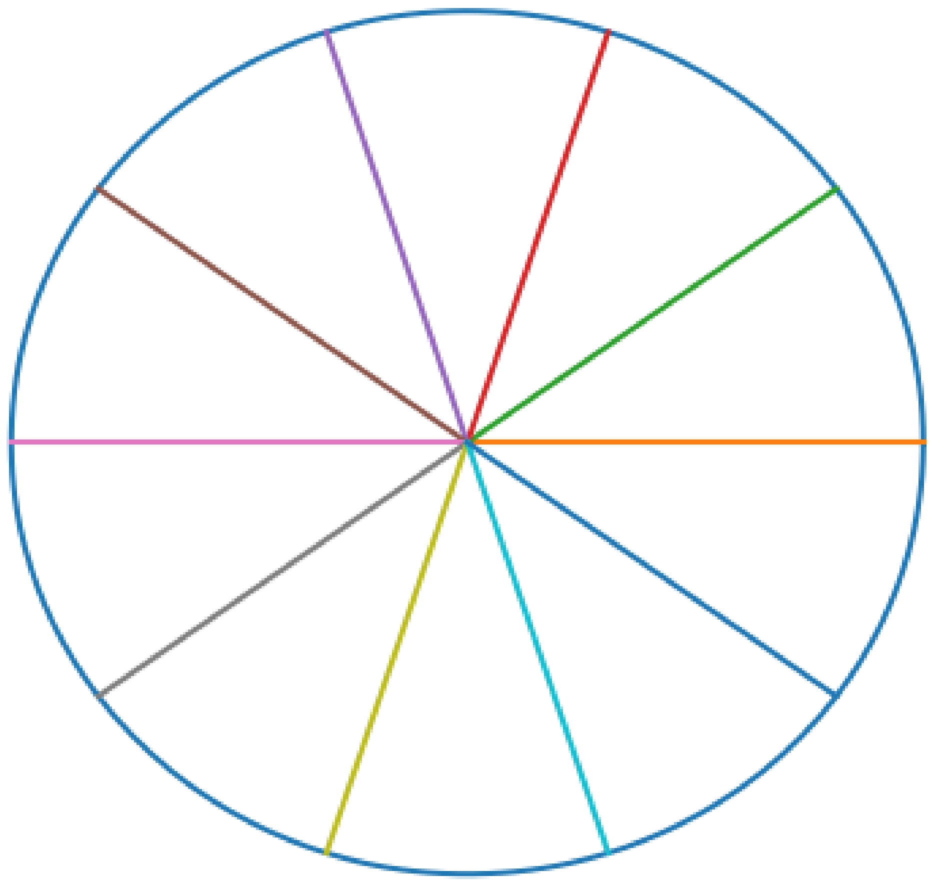
Modern artificial intelligence systems are extremely powerful, but they are also expensive to run. Large models require enormous hardware resources because traditional neural networks perform computation uniformly, even when much of it is unnecessary. Researchers are exploring ways to make AI computation sparse—activating only the parts of a model that are needed for a given input.

1. The Core Idea

Angular Manifold Routing proposes a geometric routing mechanism. When token embeddings are normalized so that they lie on a unit hypersphere, routing can be performed purely by direction rather than magnitude. This means tokens that point in similar directions share compute resources.

Mathematically, embeddings are normalized as: $$ \tilde{x} = \frac{x}{||x||_2} $$ This places all embeddings on the hypersphere S^{d-1}.

Angular sectors on a sphere (2D analogy)



2. Angular Sector Routing

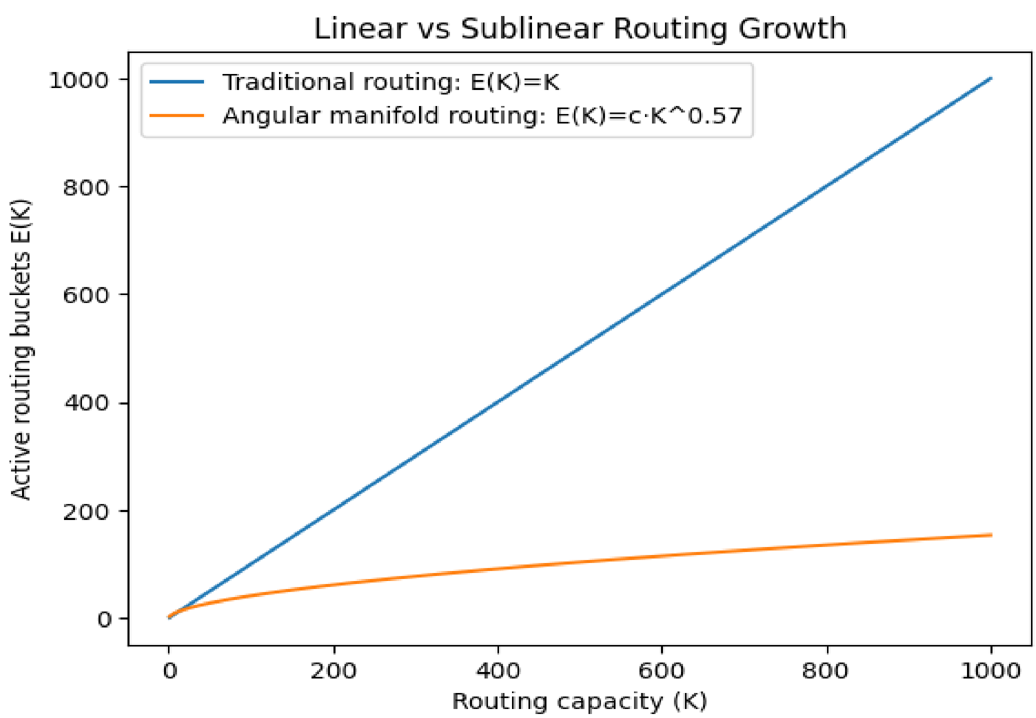
The hypersphere is partitioned into angular sectors. Each sector acts as a routing bucket. Tokens are assigned to the sector whose direction most closely matches their embedding.

Because semantically related tokens often occupy nearby directions in embedding space, tokens naturally cluster into a limited set of sectors.

3. The Key Observation

If the system provides K routing buckets, one might expect roughly K buckets to be used. However experiments consistently show that the number of active buckets grows sublinearly.

The empirical law observed is: $E(K) \approx c \cdot K^{0.57}$ where: K = number of routing sectors available E(K) = number of sectors actually used



4. Why This Matters

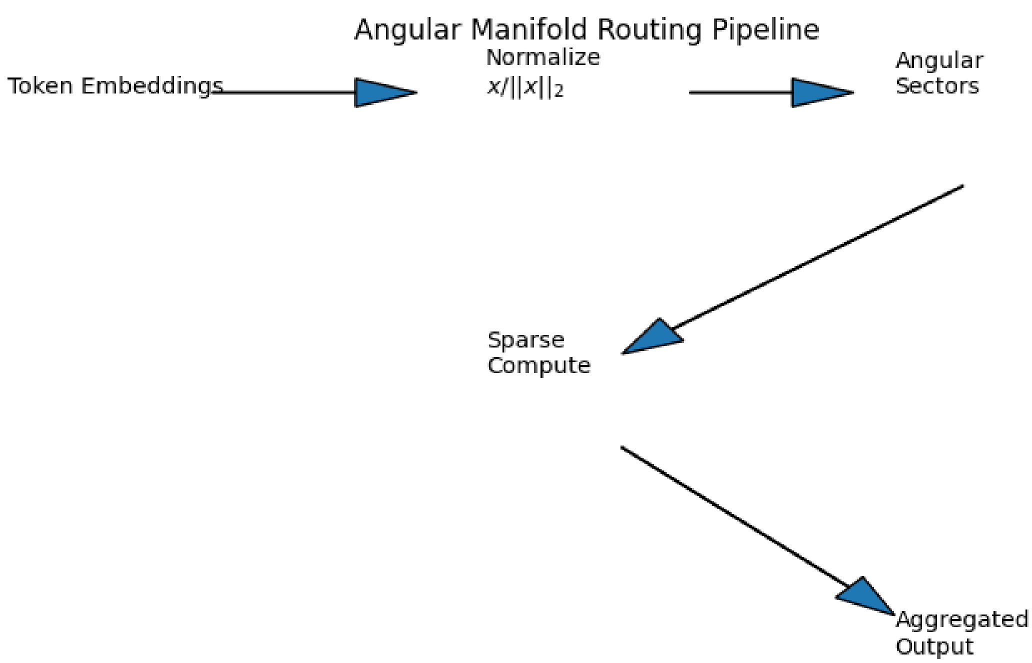
Sublinear growth means that increasing routing capacity does not increase compute linearly. Instead, the model naturally concentrates tokens into fewer buckets. This produces several advantages:

Benefit	Impact
Reduced computation	Only active buckets perform compute
Hardware efficiency	Better memory locality
Cache performance	Fewer cache misses
Cost reduction	3–5× lower effective routing cost in tests

Importantly, this effect appears to arise from geometry rather than complex learned gating networks. The embedding manifold itself organizes computation.

5. Architecture Overview

The resulting routing pipeline is relatively simple: 1. Embed tokens into vectors 2. Normalize embeddings onto the unit hypersphere 3. Partition directions into angular sectors 4. Route tokens according to direction 5. Activate compute only in occupied sectors 6. Aggregate outputs



6. Experimental Findings

Experiments tested several properties of the routing system including normalization methods, routing capacities from K = 25 to K = 1000, randomized controls, and hardware proxy models. Across these tests the scaling exponent remained consistently sublinear, typically between 0.50 and 0.64 during training.

7. What Did Not Matter

Two factors surprisingly had little impact: • Vector norm choice (L1, L2, L3, L4) • Radial shell partitions These results suggest the effect is driven primarily by angular geometry rather than radial structure.

8. Implications for AI Systems

If validated in full production models, Angular Manifold Routing could enable more scalable AI architectures by concentrating computation into fewer active experts while maintaining high model capacity.

Potential benefits include faster inference, reduced energy consumption, lower infrastructure costs, and more efficient mixture■of■experts systems.

Summary

Routing tokens by direction on a normalized hypersphere causes them to cluster into far fewer compute buckets than expected. This geometric property creates natural sparsity and could significantly improve the efficiency of large■scale AI models.